

Ethan Gates

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SUMMARY

AI-Native Software Engineer building production AI products, agent systems, and data-intensive full-stack applications. Healthcare SaaS engineer with hands-on work in grounded AI, local inference, orchestration, and evaluation.

EDUCATION

B.S. Computer Science, Western Governors University

Completed: Feb 2025

PROFESSIONAL EXPERIENCE

Software Engineer, DOVAXIS (Healthcare SaaS patient referrals & admissions)

May 2025 – Present

Began as a paid engineering intern; converted to full-time November 2025.

- Owned a multi-tenant authorization migration end to end: consolidated scattered role, route, and resource checks into a Permit.io-backed facade with bulk decisions and cached permissions, built the reconciliation report and per-organization readiness verdict that gated the cutover, then ran the production cutover of every customer organization over two days.
- Built and operate a standalone multi-provider LLM service (GraphQL on Cloud Run) that internal services call instead of the OpenAI SDK: a model registry with provider auto-detection across OpenAI and Gemini, a unified quality tier, streaming and tool calling, per-call telemetry for tokens, USD cost, latency, and organization, and later a gated public chat with web search and cited sources.
- Kept a multi-tenant clinical document library and its AI search index in agreement: unified the version-history and DocHub lifecycle, fixed a defect that left superseded forms in the vector index after a replacement, surfaced failed re-indexes, and restored cited-source attribution so answers separate the documents they used from other hits.
- Rewrote the regulation library as a multi-tenant system of record: a master library client organizations subscribe to, a diff-and-sync service that flags new, updated, or removed items while preserving client customizations, read-only import into each client's document tree, and selected regulations wired into vector search so client AI answers cite them.
- Rebuilt how a legacy healthcare app proves changes are safe, using agentic testing workflows to generate and govern the suites: an offline Firebase emulator stack with sanitized fixtures that moved data-writing tests off production, browser and Firebase-backed suites on every pull request, 1,156 Jest checks derived from twenty real production bug fixes, and Playwright governance through five static gates and a coverage ratchet.

INDEPENDENT PROJECTS

- **GatesAI Chat**: Built and shipped a local-first AI workspace (React/Tauri/Rust with a Go bridge): OpenRouter/Ollama routing, multi-round tool execution, sandboxed files/shell/Git/SQlite, memory, and image generation, with automated checks for streams, tools, persistence, and 100+ user journeys. [\[Repo\]](#)
- **Fathom**: Built a market-analysis suite in TypeScript and React: portfolio backtesting over 155 years of asset-class history, Trinity-validated Monte Carlo retirement simulation, SEC-filing research, and broker-CSV portfolio X-ray. [\[Demo\]](#)
- **Neon Vector Defense**: Shipped a hosted React/TypeScript Canvas tower-defense game with deterministic simulation, bot playtests, headless balance harnesses, compact replay verification, Firebase leaderboards, and server-side anti-cheat. [\[Demo\]](#)
- **Media Vault**: Built a private photo and video library that runs on your own machine: timeline, people, places, trips, and search, with local models for faces, text in photos, image search by description, and video transcripts. [\[Repo\]](#)
- **Milo**: Built a local voice companion: whisper.cpp, Ollama, and Pocket TTS in one Python process, interruptible mid-sentence, first audio about 300 ms after you stop talking, and an animated robot whose mouth follows playback amplitude. [\[Repo\]](#)

TECHNICAL SKILLS

- **Languages**: TypeScript, Python, Rust, JavaScript, SQL, C#.
- **AI / LLM Engineering**: Agent orchestration, context management, tool calling, grounded retrieval, embeddings and vector search, evaluation and benchmarking, cost and latency telemetry, OpenAI, Gemini, OpenRouter, Ollama.
- **Frameworks & Libraries**: React, MobX, FastAPI, Tauri, TensorFlow, NumPy, Pandas, Tailwind CSS.
- **Tools & Infrastructure**: Git/GitHub, Playwright, Firebase, PostgreSQL, Docker, Vite, Cloud Run, Linux.